

Fynla — Tech Stack and High-Level Architecture

Stamp: commit [a7c13608a](#), branch [fix/rsu-value-labels-free-tier-capture](#), 2026-09-22. **Method:** every claim below was read from a file in this run. Where I could not read the backing evidence it says "I COULD NOT VERIFY".

1. In plain English

Fynla is one Laravel application with one MySQL database, served to three front ends: the desktop web app, the [/m](#) mobile web app, and a native iPhone app. All three talk to the same HTTP API. Behind the API, each of the seven planning modules has a "module agent" that runs the calculations and produces recommendations, and Fyn, the AI companion, sits on top of those same engines. It is hosted on shared SiteGround hosting, built locally, and deployed by hand.

2. Runtime stack

Layer	Technology	Evidence
Language	PHP ^8.2 (local machine runs 8.5.2)	composer.json require; <code>php -v</code>
Backend framework	Laravel ^10.10	composer.json
Auth	Laravel Sanctum ^3.3 (bearer tokens; 61 auth:sanctum groups in routes/api.php) plus TOTP MFA via pragmarx/google2fa-laravel	composer.json ; routes/api.php:186,272,279
Database	MySQL 8 (DB_CONNECTION default <code>mysql</code>); 342 migrations; 125 model files	config/database.php:19 ; <code>ls database/migrations</code> ; <code>ls app/Models</code>
Cache / session / queue	Defaults file / file / sync in the env template; queue:restart is part of the csjones deploy runbook, so a worker is expected on servers. I COULD NOT VERIFY the live .env values.	.env.example:24–28 ; deploy/DEPLOY.md step 4
Mail	MAIL_MAILER default log ; template says smtp	config/mail.php:16 ; .env.example:37
File storage	local default, S3 disk configured	config/filesystems.php:16,47
Frontend framework	Vue 3 ^3.5 , Vue Router 4, Vuex 4 (+ vuex-persistedstate)	package.json
Build	Vite 5 with laravel-vite-plugin , vite-plugin-pwa (web only); separate vite.mobile.config.js for /m with no Laravel plugin and no PWA	vite.config.js ; vite.mobile.config.js
Styling	Tailwind CSS 3 with the Fynla palette tokens (raspberry , horizon , spring , violet , savannah , eggshell)	tailwind.config.js:15–48
Charts	ApexCharts 5 via vue3-apexcharts	package.json
Rich text	TipTap 3 (editor extensions for tables, images, links)	package.json
Native iOS	SwiftUI, Xcode project ios-native/Fynla.xcodeproj , two schemes (Fynla-Staging , Fynla-Production), both pointing at fynla.org	ios-native/CLAUDE.md
PDF / documents	barryvdh/laravel-dompdf , phpoffice/phpspreadsheet , phpoffice/phpword , smalot/pdfparser , intervention/image , mews/purifier	composer.json
AI SDKs	anthropic-ai/sdk , openai-php/client (composer); plus a hand-written XaiClient	composer.json ; app/Services/AI/XaiClient.php

3. Third-party services

Service	Used for	Evidence
xAI (grok-4.3 chat, vision and "advanced" models)	Fyn chat and document extraction when AI_PROVIDER=xai ; runtime override stored in cache from the admin panel	config/services.php:40-53 ; app/Services/Documents/AIExtractionService.php:127-189
Anthropic (claude-haiku-4-5 , claude-sonnet-4-6)	The other chat provider; also the marketing pipeline's AnthropicOpusClient	config/services.php:34-38 ; app/Services/Pipeline/AnthropicOpusClient.php
Revolut Merchant	Subscription payments (sandbox on dev, live on prod); tier price sync	config/services.php:60-65 ; app/Services/Payment/Revolut*.php ; app/Services/Tiers/RevolutTierVariationSync.php
Apple App Store Server API	Native in-app purchase verification through a Python bridge (services/apple_store_bridge/cli.py)	config/apple_store.php:18-39
APNs / FCM	Push notifications to native and mobile	config/services.php:67-81 ; app/Services/Mobile/ApnsClient.php
getAddress.io	Postcode lookup	config/services.php:56
Companies House	Filing due dates for a user's limited company	config/services.php:100-104
GitHub Issues API	In-app bug reports become issues, feeding the claude.yml auto-fix workflow	config/services.php:83-89 ; .github/workflows/claude.yml
Pexels	Stock cover images for CMS articles	config/services.php:92-95
Plausible	Privacy-first web analytics	config/analytics.php:30
Awin	Affiliate attribution (off by default, on in the prod template)	workforce/core/registry/capabilities.md §7.3
HeyGen, local Whisper, FFmpeg	Marketing video pipeline: script -> avatar video -> transcription -> clips	app/Services/HeyGenService.php:48 ; app/Services/Pipeline/LocalWhisperTranscriber.php ; app/Services/FFmpegService.php
Google Drive (service account)	Word documents in Drive become InsightArticle records	app/Services/Pipeline/Google/ ; workforce/core/registry/storage.md §3
AWS S3	Application file storage	config/filesystems.php:47 ; storage.md §3

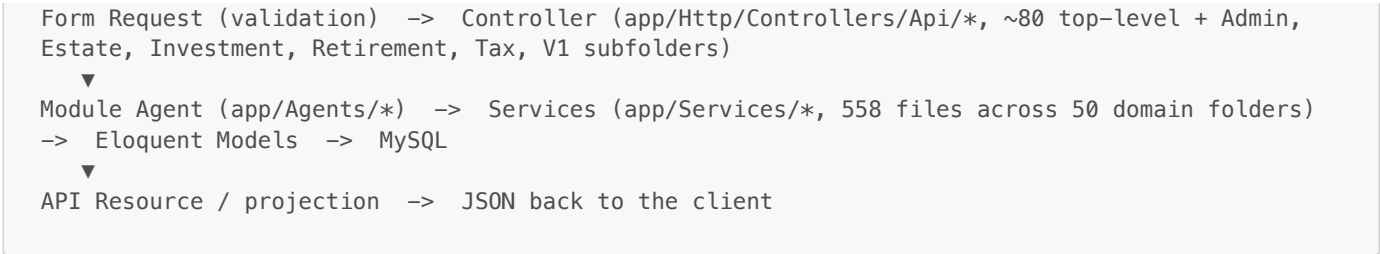
4. High-level architecture

4.1 Request flow (all three clients)

```

Browser / iPhone
|   HTTPS, Sanctum bearer token
▼
routes/api.php (1,577 lines) · routes/api_v1.php (native /api/v1/native/*) · routes/web.php
(server-rendered public pages + SPA catch-all)
▼

```



Evidence: `CLAUDE.md` flow line; `ls app/Http/Controllers/Api; ls app/Services; wc -l routes/*.php`.

4.2 The three clients

Client	Source	Bundle	Notes
Desktop web SPA	<code>resources/js/</code> (36 Vuex modules, views per module)	<code>public/build/</code> via <code>vite.config.js</code>	PWA plugin enabled; router base and API base set by the per-environment build script
<code>/m</code> mobile web	<code>resources/mobile/</code> (own <code>api.js</code> , <code>router.js</code> , <code>store.js</code> , <code>tokens.js</code>)	<code>public/m-build/</code> via <code>vite.mobile.config.js</code>	Deliberately isolated: no Laravel Vite plugin, no PWA, <code>publicDir: false</code> ; only shared file is <code>store/modules/auth.js</code>
Native iOS	<code>ios-native/</code> SwiftUI	Xcode archive via <code>deploy/mobile-native/</code>	Layers: <code>Core/API</code> , <code>Core/Authentication</code> , <code>Core/Keychain</code> , <code>Core/Biometrics</code> , <code>Core/StoreKit</code> , <code>Core/DesignSystem</code> , <code>Features/{Area}</code> ; decodes with <code>TolerantDecoding</code> ; native-only session routes under <code>/api/v1/native/auth/session/*</code>

Evidence: `vite.config.js`; `vite.mobile.config.js`; `ios-native/CLAUDE.md` "Layering"; `routes/api_v1.php:48–118`.

4.3 Public pages are not the SPA

The homepage, `/how-it-works`, `/features`, `/pricing` and the other marketing pages are server-rendered PHP files under `public/pages/`, included by closures in `routes/web.php` and cached for five minutes. Authenticated users are redirected to the SPA by the `redirect.authed` middleware. This is why compiling the route cache is banned on this app: the compiled matcher lets the SPA catch-all shadow `/`.

Evidence: `routes/web.php:96–135`; `deploy/DEPLOY.md` "NEVER php artisan optimize".

4.4 The AI layer (summary; full detail in `agentsMap.md`)

- One endpoint for every surface: `POST /api/ai-chat/conversations/{id}/messages`, handled by `AiChatController`.
- One unified system prompt (`FynSystemPrompt` + `FynContextAssembler`), two write states enforced by tool-gating (Onboarding `Fyn` writes, Advice `Fyn` is read-only).
- A CoALA decision loop under `app/Services/AI/Loop/` with a cycle cap of 8 per turn, and three durable memory stores as markdown under `fyn-memory/` (procedural, episodic, semantic) plus per-user semantic memory under `storage/app/memory/`.
- Provider switchable between Anthropic and xAI; per-user weekly token budgets live in `tier_configurations`.

Evidence: `.claude/skills/fyn-architecture/SKILL.md`; `config/fyn.php:17–68`; `fyn-memory/README.md`; `workforce/core/constitution/06-commercials.md` §3.

4.5 Commercial and entitlement layer

Two tiers only, Free (£0) and Premium (£6.99 monthly / £59.99 annual), seeded into `tier_configurations` by `TierConfigurationSeeder`. `TierResolver` and `PremiumEntitlementResolver` decide a user's tier; `DbTierGate` and `TeaserGate` enforce it. Free tier has count caps (for example 2 savings, 2 investments, 2 pensions, 1 property) and the Estate module in "teaser" state. Legacy paid plans are grandfathered.

Evidence: `database/seeder/TierConfigurationSeeder.php:31-74`; `app/Services/Tiers/workforce/core/constitution/06-commercials.md §1-2`.

4.6 Background work

The Laravel scheduler (`app/Console/Kernel.php`) runs 41 entries: subscription expiry and renewal reminders, account deletion and retention purges, module alert emails (protection, savings, estate, mortgage rates, business filings), the lifecycle email engine at 08:30, daily insight push at 08:00, payment reconciliation every ten minutes, AI audit retention and chain verification weekly, stale-conversation summarisation every thirty minutes, episodic memory reconcile nightly and cold-archive weekly, and the marketing pipeline (Drive watch, post scheduling, weekly social report). Queue jobs include `ConversationSummariserJob`, `RunMonteCarloSimulation`, `RecalculateRiskProfileJob`, `ProcessAppleNotification`, `FireAwinConversionJob` and `PublishScheduledInsightsJob`.

Evidence: `app/Console/Kernel.php:21-111`; `ls app/Jobs`.

4.7 Data protection and audit

- Encryption of existing data is a one-off artisan command (`data:encrypt`, `app/Console/Commands/EncryptExistingData.php:27`). **I COULD NOT VERIFY** which model attributes are cast encrypted: a grep for the `encrypted` cast across `app/Models` returned zero files in this run.
- GDPR: `GDPRController`, scheduled and grace deletions, retention warnings and purges, `FynUserErase` for AI memory (`app/Console/Commands/Kernel.php:22-27`).
- AI audit chain: `AuditChainService`, `AiAuditRetentionJob`, `ai:audit:verify-chain` weekly (`Kernel.php:59-60`).
- Preview personas are isolated by `is_preview_user` and `PreviewWriteInterceptor` (`CLAUDE.md` Rule 1 and 7).

5. Environments and deployment

	Production	Dev / staging
URL	fynla.org	csjones.co/fynla
Branch	main	dev
Hosting	SiteGround shared hosting (no <code>DirectoryMatch</code> , memory too low for npm)	same
Deploy	manual upload; <code>ssh-fynla</code> MCP is production-only	<code>git pull origin dev</code> + upload <code>public/build/</code> and <code>public/m-build/</code>
Build	<code>./deploy/fynla-org/build.sh</code> (<code>VITE_BASE_PATH=/build/</code>)	<code>./deploy/csjones-fynla/build.sh</code> (<code>/fynla/build/</code>)

Evidence: `deploy/DEPLOY.md:1-80`; `deploy/README.md:25,101`; `workforce/core/registry/systems.md §2`.

Flow: feature branch -> PR to `dev` -> PR `dev` -> `main`. Nothing reaches `main` without going through `dev`.

6. Quality tooling and CI

Tool	Where
Pest 2 (486 Unit, 556 Feature, 4 Integration, 27 Browser files; 24 BS-NN scenarios)	<code>tests/</code> ; <code>.github/workflows/quality.yml</code> runs per suite
Vitest 3 + <code>@vue/test-utils</code>	<code>tests/frontend</code> , <code>resources/**/*.__tests__</code>
Playwright 1.56 (smoke + full E2E; nightly on chrome and webkit)	<code>tests/E2E/*.spec.js</code> ; <code>.github/workflows/nightly.yml</code>
ESLint 9, Pint (PSR-12), policy lint, mobile-impact check	<code>package.json</code> scripts; <code>scripts/quality/</code>
Claude Code hooks: dangerous-command guard, env guard, prod guard, tax-hardcode check, design lint, <code>/m</code> parity check, oversight and workforce guards,	<code>.claude/hooks/</code> ; <code>.claude/settings.json</code>

Tool	Where
pre-compact handover	
CI workflows: Quality Gate, Nightly Quality (E2E + audits), iOS Native (macOS runner, <code>xcodebuild</code>), Logic Guard (dashboard/onboarding), Main Branch Guard, Claude Auto Bug Fix	<code>.github/workflows/</code>
Fyn eval harness (HTTP-driven, SSE consumer, recall floor 95 per module)	<code>app/Services/Eval/;</code> <code>config/fyn_eval.php:22-33</code>
Local MCP servers: <code>mysql</code> , <code>playwright</code> , <code>ssh-fynla</code>	<code>.mcp.json</code>

7. Where the knowledge lives

- `CLAUDE.md` at the root and nested in `app/Http/`, `app/Services/`, `database/`, `tests/`, `resources/js/`, `ios-native/`.
- `.claude/skills/` (workflow skills such as `fyn-architecture`, `verify-m`, `release`, `app-map`) and `.claude/agents/` (the workforce, see `agentsMap.md`).
- `workforce/core/` (constitution, charter, registries), `workforce/ops/` (board of 347 items, gates, missions, event log).
- `fynlaBrain/` Obsidian vault on the desktop (mirror, not source).
- `docs/app-map/` section maps and `docs/diagrams/` Excalidraw.